

Questions for Julian Schumann, conference meeting, September 2026

Private preparation for Jonas; not to be handed over. The paper handout for Julian is docs/handout_schumann_2026-09.md (PDF beside it); its section 6 carries eight of these questions in collegial form, marked [H1]–[H8] below. Everything else here is for asking verbally. It draws on the authors' reply of 2026-09-11 (correspondence/2026-09-11_authors_reply_to_method_review.md), which was written by Julian's co-author on behalf of both.

If there are only fifteen minutes

- 1 **The road departures — his reading.** The co-author's letter said our interpretation was one-sided and offered to explain at the conference. Asking first, and listening, is probably the best way to open.
- 2 **Comfort or collision?** Was the safety-margin counterfactual meant to describe comfort at all? [H1]
- 3 **Where a driver trait would live in the model.** [H3]
- 4 **The gate from inside the model** — the norm tournament's collapse of trust as the onset. [H5]
- 5 **Interest in a joint look at cut-in data.** [H8]

Opening

- Thank him for the reply and for the correction he is preparing. The letter came from his co-author; he may not have seen its exact wording, so the safest reference is to the substance ("we heard that the issues were reporting or undiscussed simplifications, and that our reading of the road departures was one-sided").
- The handout can be given at the start; it sets out what we did so the conversation can go to the open issues.

Follow-ups to our review and his reply (verbal only)

- a. **Road departures at 25 m/s.** Our reading is in docs/method_review.md §4.1 (closing sentence marked opinion) and the authors' edition, chapter 05. What is the other side: intended behavior, a known limitation of the lane-change control at speed, or something in how departures are classified?
- b. **The mislabeled scenario in a figure caption.** Does it account for the Fig. 3 worked example not matching the deposit, and for the Fig. 3b explanation reading reversed (method_review.md §4.3)? That is our guess, not something they said.
- c. **The correction.** Timeline, and what it covers, so that we can add dated notes to our documents when it is out (the list is in the filed reply).
- d. **Which is "the model" where the code and the SI differ?** For example the one-sided inverse-tau term, the total-acceleration control-effort term, and the off-road cost of $-15\,000$ against the SI's $-5\,000$ (method_review.md §5). The reply calls the SI issues typos, which suggests the code is the reference; worth confirming, since our mirror follows the code.
- e. **The accumulator's starting point** is, in their words, a simplification. What would he use for continuous driving? [H7]
- f. **Perception noise above the looming threshold** is about 10^{-5} in the released configuration, which makes perception effectively exact once the threshold is crossed (method_review.md §6.3). Deliberate? [H6]

Comfort-zone boundaries

- 1 **Comfort or collision** [H1]. *Why we ask:* our pre-registered test ruled the deficit out as a comfort measure on the second cut-in study, and about half of that loss is the worst-case counterfactual (docs/r2_pipeline_review.md §3.2–3.3). If the term was only ever meant for collision avoidance, the result is not a criticism of it, and a comfort reference is a separate construction.
- 2 **The inverse-tau preference** [H2]. *Corrected 2026-09-12:* the mean of 0.2 s^{-1} is sourced. SI §2.4, just after Eq. 48, takes it from Markkula et al. (2016), the "Farewell" paper: in 116 crashes and 241 near-crashes, few drivers braked before looming reached 0.2 s^{-1} and most within a second after, with the same cut-off at $\theta \approx 0.02\text{ rad/s}$. So do not ask where it comes from. Ask instead: (a) turning an emergency brake-onset level into the centre of a *preference*

reads like a comfort standard — was that intended? (b) the spread of 0.125 s^{-1} has no stated source; the code derives it from the collision cost, $0.25 / (\log_{10} 10\,000 - 2)$. (c) Farewell reads responsiveness to looming as driver- and situation-dependent (expectancy, driving style, drowsiness, visibility), which makes a per-driver τ^{-1} mean a candidate home for our trait (question 3). *Talking point, verbal only*: our video cut-in level has a population median of 0.032 rad/s (out/stage1_looming.md), the same variable as Farewell's 0.02 rad/s cut-off and of similar order, though the scenario and the response differ.

- 3 **Where a stable trait lives** [H3]. We find one per-driver level shared across four scenarios at about 69% of the reliability ceiling (out/cross_scenario_consistency.md), and fitted cut-in and left-turn levels agree at $+0.65$ (out/driver_levels.md). Which parameter would he map that onto: t_{react} , the assumed worst-case braking, λ , a tolerance?
- 4 **Calibrating the assumed worst-case braking**. The free-following table covers steady headways only up to about $1.0\text{--}2.1 \text{ s}$ depending on speed (method_review.md §6.2). Would naturalistic following headways be the natural calibration source?
- 5 **Looming as the cue**. Our video data prefer the expansion rate; Xue et al. (2018) found inverse tau better for brake onsets in a simulator. Does he have a view on which, and on accumulating looming rather than a threshold on it?
- 6 **Surprise as the onset** [H5]. docs/surprise_without_the_field.md argues that world surprise could define when a situation starts to count. His norm tournament already computes a version of that (the candidate-independent "now" weight collapsing). Would he use it that way?

Cut-in

- 1 **Norms for a vehicle partway into the lane** [H4]: a progress-dependent norm, since a lane change is legitimate but only for a plausible duration.
- 2 **The binary lateral tests** (3 widths for perception, 1.15 for the preference terms). Did they try a graded version, and does the oncoming incursion avoid the issue only because it crosses quickly?
- 3 **The steering-noise dial**. $w_{\text{sd_model}}$ is 0.0045 in the rear-end scenario and 0.4575 in the two lateral ones, a factor of 100 (method_review.md §5 item 4, §9 question 4). Is the rear-end value intentional, and would he start a cut-in from the lateral value?
- 4 **Their own plans or data**. Do they (TU Delft, Waymo) have human cut-in response data, or a cut-in scenario planned? Would a cut-in make a sensible second held-out test?
- 5 **Running the loop**. What does a scenario cost them on GPU, and is there a faster implementation? [H8]

Crash causation

- 1 **Beliefs coasting through a glance** [H6]. Intended? And why the planner's hard-coded "avoid off gaze" in this paper: only to isolate avoidance behavior?
- 2 **Glance choice by epistemic value**. Is the machinery from Engström et al. (2024) ready to switch on ($\text{road_gaze_preference} \neq 0$) in this code base, or would he advise against it?
- 3 **The accumulation rate**. $\lambda = 10^{-5.95}$ in the released rear-end setup. Which human data set it? Our onsets came out slower and more variable than the fixed-delay rule (median 1.25 s against 0.50 s after the anchor); does that match his expectation?
- 4 **The pointwise surrogate** [H8]. Our open-loop approximation matched the closed loop to a median 0.55 s over 23 scenarios. Would he accept it for population-scale studies?
- 5 **Reactive other agents**. Any plans for other road users that respond to the driver?

What we could offer

- The surprise-measure library, the NumPy mirror of the preference terms (following the code), and the OSF validation harness.
- A joint analysis of the cut-in data, if the data owners agree; the two video studies come from a related project and are not ours to share.

Worth avoiding

- Re-arguing the review. The numbers are not in dispute and the interpretation is what they offered to discuss; the aim is to hear it.